Almost independent distributions separated by logarithmic-width DNFs

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Allowed standard prerequisites (editorial).

- Standard probability, circuits and finite-group representation background. All author local support retained.

1 Introduction

Linial and Nisan [16] conjectured that "k-wise independent" distributions fool constant-depth circuits (class AC^0). More specifically, a distribution $\mathcal{D}$ over $\{0,1\}^n$ is called *k-independent* if the marginal distribution on every k-sized subset of bits is uniform. We say that $\mathcal{D}$ *δ-fools* a circuit C if the circuit cannot distinguish $\mathcal{D}$ from the uniform distribution on $\{0,1\}^n$:

$$\left| \Pr_{\boldsymbol{x}\sim\mathcal{D}}[C(\boldsymbol{x})=1] - \Pr_{\boldsymbol{x}\sim\{0,1\}^n}[C(\boldsymbol{x})=1] \right| \;\le\; \delta.$$

The Linial–Nisan conjecture was first proved for depth-2 circuits (DNFs and CNFs) by Bazzi [4] (with a simplification by Razborov [21]) and then for every AC^0-circuit by Braverman [7]. Indeed, Braverman showed that every size-s AC^0-circuit is $o(1)$-fooled by $\mathrm{poly}(\log s)$-independence.

Aaronson [1] asked whether the Linial–Nisan conjecture could be strengthened to hold also for "almost k-wise independence," a seemingly modest generalisation. We say that a distribution $\mathcal{D}$ over $\{0,1\}^n$ is *(ε,k)-independent* if for every subset $I\subseteq[n]$, $|I|=k$, the marginal distribution on the bits in I is multiplicatively close to uniform in the sense that for every $\alpha\in\{0,1\}^I$,

$$(1-\varepsilon)2^{-k} \;\le\; \Pr_{\boldsymbol{x}\sim\mathcal{D}}[\boldsymbol{x}_I=\alpha] \;\le\; (1+\varepsilon)2^{-k}.$$

Generalised Linial–Nisan Conjecture (GLN).
Let $\mathcal{D}$ be a $(1/n^{\Omega(1)}, n^{\Omega(1)})$-independent distribution over $\{0,1\}^n$. Then $\mathcal{D}$ $o(1)$-fools every AC^0-circuit of size $2^{n^{o(1)}}$.

Aaronson's original motivation for this conjecture was to resolve a problem in quantum complexity theory. He showed that a positive resolution of GLN would imply the separation $\mathsf{BQP}\not\subseteq\mathsf{PH}$ relative to an oracle. (This separation was subsequently proved by Raz and Tal [20] by a different approach.) Later, Aaronson himself found a counterexample to GLN for depth-3 circuits [2], but he still re-posed the conjecture (and thought it "plausible") for depth-2 circuits. Our main result here is to refute the GLN conjecture in this remaining case.

Theorem 1 (Main result).
There exists a $(1/n^{\Omega(1)}, n^{\Omega(1)})$-independent distribution $\mathcal{D}$ over $\{0,1\}^n$ and a $O(\log^3 n)$-width DNF formula F such that

$$\Pr_{\boldsymbol{x}\sim\mathcal{D}}[F(\boldsymbol{x})=1] - \Pr_{\boldsymbol{x}\sim\{0,1\}^n}[F(\boldsymbol{x})=1] \;\ge\; \Omega(1).$$

Let us make two notes about the parameters here. First, our formula F will have quasi-polynomial size, whereas Aaronson's depth-3 counterexample has only polynomial size; hence his example achieves slightly better parameters (at the cost of larger depth). Second, our construction can be varied to produce the following tradeoff: by increasing the DNF width to any $w \leq n^{o(1)}$, we can make the distribution $(\exp(-w^{\Omega(1)}), n^{\Omega(1)})$-independent (see Section 2.5).

1.1 Implications and related work

One consequence of the failure of the GLN conjecture is to the construction of *pseudorandom generators* (PRGs) for DNFs. It is known that for $k \geq \Omega(\log n)$ there exist $(o(1), k)$-independent distributions with support size $2^{O(k)}$ [18, 3], which is smaller than $n^{\Omega(k)}$ that is required for truly k-wise independent distributions [8]. Thus Theorem 1 rules out a natural approach ("output an almost k-wise independent distribution") to improving the seed length of PRGs. For the current state-of-the-art PRGs for DNFs, see [9, 22, 17]; see also the survey [13].

Another lesson from Theorem 1 is to the further development of circuit lower bound methods. We find it important to seek alternative proofs of central theorems such as Bazzi's [4, 21] and its extensions [7]. The existing proofs use the *polynomial method* to approximate a DNF with a low-degree polynomial. Is there a more "combinatorial" proof of Bazzi's theorem? One such more combinatorial approach is the *top-down* lower bound method [12, 11], which often uses entropy-based arguments to analyse circuits. We interpret the failure of GLN as a challenge to such top-down methods. While the method is in a formal sense *complete* (it can prove any lower bound that is true), the typical entropy counting arguments have a hard time distinguishing almost k-wise independent distributions from truly k-wise independent ones, suggesting that any top-down proof of Bazzi's theorem would require substantially new ideas.

Finally, we mention that – besides (almost) k-wise independence – several other notions of pseudorandomness have been considered in the literature [6, 5, 14].

1.2 Workaround: Local couplings

To complement our main result, we also propose a way to circumvent the failure of GLN by proposing a new notion of pseudorandomness called *local couplings*. This notion is useful for fooling depth-2 circuit models, but not depth-3 models; in particular, we show the following claims:

1.

Local couplings fool DNFs (query complexity analogue of NP).

2.

Local couplings fool decision lists (query complexity analogue of P^{NP}).

3.

Local couplings do *not* fool depth-3 circuits (query complexity analogue of $\Sigma_2\mathsf{P}$).

Definition 2 (Local couplings).
A pair of jointly distributed random variables $(\boldsymbol{x}, \boldsymbol{y}) \in (\{0,1\}^n)^2$ is an ε-semi-coupling if for every $y \in \operatorname{supp}(\boldsymbol{y})$ and $i \in [n]$,

$$\Pr[\boldsymbol{x}_i \neq \boldsymbol{y}_i \mid \boldsymbol{y} = y] \leq \varepsilon.$$

We say that $(\boldsymbol{x}, \boldsymbol{y})$ is an ε-coupling if both $(\boldsymbol{x}, \boldsymbol{y})$ and $(\boldsymbol{y}, \boldsymbol{x})$ are ε-semi-couplings.

The notion of a local coupling was somewhat implicit in Aaronson's analysis [2] of his depth-3 counterexample. Local couplings are also a stronger variant of a notion proposed by Zhandry [23] that he called "substitution distance."

Claims 1–2

A width-k *decision list* is a sequence of pairs $\{(T_i, a_i)\}_{i \in [m]}$ where T_i are k-terms (conjunctions of at most k literals) and $a_i \in \{0,1\}$ are output values. A decision list defines $f\colon \{0,1\}^n \to \{0,1\}$ as follows: $f(x) = a_i$ where $i = \min\{i \in [m] \mid T_i(x) = 1\}$. The decision list width of a function is polynomially equivalent to the number of DNF queries necessary to compute the function [10, Appendix A]. In other words it is indeed a query complexity analogue of P^{NP}. The following theorem (Section 3.1) formalises 1–2 when $\boldsymbol{y}$ is uniformly distributed.

Theorem 3.
Let f be computed by a width-k decision list. For any ε-coupling $(\boldsymbol{x}, \boldsymbol{y})$,

$$\Pr[f(\boldsymbol{x}) \neq f(\boldsymbol{y})] \leq 2k\varepsilon.$$

Claim 3

Aaronson's [2] original counterexample involved a distribution $\mathcal{D}$ related to a certain *surjectivity* function, which can be computed by a small depth-3 circuit. We observe (Section 3.2) that Aaronson's distribution $\mathcal{D}$ can indeed be locally coupled with the uniform distribution, which implies that local couplings do not fool depth-3 circuits (Claim 3). We can furthermore conclude (using Theorem 3) that $\mathcal{D}$ fools decision lists – this claim was already made earlier by Aaronson [2, Theorem 3], but his proof contained a mistake,[1] which we can now fix with the notion of local couplings. Finally, we also show (Section 3.3) that an ε-semi-coupling is not enough by itself to fool DNFs – one truly needs the two-sided condition of an ε-coupling.

2 Counterexample

In this section, we prove Theorem 1 by constructing a DNF formula F and an associated almost k-independent distribution $\mathcal{D}$ that F can distinguish from uniform. We first (Section 2.1) construct a weak example that distinguishes $\mathcal{D}$ from uniform with advantage $1/\mathrm{poly}(\log n)$. Then (Section 2.4) we amplify this advantage to $\Omega(1)$ by using a standard majority trick.

2.1 Construction

Our starting point is the *address* function $\textsc{Addr}\colon \{0,1\}^m \times \{0,1\}^{2^m} \to \{0,1\}$ defined as $\textsc{Addr}(a,p) := p_a$. Here we write p_a to mean $p_{\mathrm{int}(a)}$ where $\mathrm{int}(a) \in [2^m]$ is the integer corresponding naturally to the bitstring a. Let us first observe that Addr together with the uniform distribution over $\textsc{Addr}^{-1}(1)$ "almost works" as the counterexample in Theorem 1. For $(\boldsymbol{a},\boldsymbol{p}) \sim \textsc{Addr}^{-1}(1)$ the distribution of $\boldsymbol{p}$ is already $(o(1), 2^{\Omega(m)})$-independent. The reason the whole $(\boldsymbol{a},\boldsymbol{p})$ does not have the same property is that for example, fixing all bits of $\boldsymbol{a}$ to some a forces $\boldsymbol{p}_a = 1$, so for some $I \subseteq [m+2^m]$ containing all bits describing $\boldsymbol{a}$ and the bit $\boldsymbol{p}_a$ the settings of $(\boldsymbol{a},\boldsymbol{p})_I$ with $\boldsymbol{p}_a = 0$ have probability zero.

To avoid this issue we hide the bits of the address using the usual *tribes* function:

$$\textsc{Tribes}(A) := \bigvee_{j\in[r]} \bigwedge_{k\in[m]} A_{k,j}.$$

The input here is an $m \times r$ boolean matrix and the function returns 1 iff the matrix contains an all-1 column. It is well-known [19, §4.2] that if we choose $r := \lceil 2^m \ln 2 \rceil$, the function becomes *balanced*, meaning that $\Pr_{\boldsymbol{A}\sim\{0,1\}^{m\times r}}[\textsc{Tribes}(\boldsymbol{A})] = 1/2 + o(1)$.

A natural attempt to define a counterexample would be to consider the distinguishing function $\textsc{Addr}((\textsc{Tribes}(A^1),\dots,\textsc{Tribes}(A^m)),p)$. This does not work since this function requires polynomial DNF width as the negation of Tribes reduces to it. We fix this by replacing the Addr function by its *monotone* version: $\textsc{mAddr}\colon \{0,1\}^m \times \{0,1\}^{2^m} \to \{0,1\}$ is defined as

$$\textsc{mAddr}(a,p) := \begin{cases} 0 & \text{if } |a| < m/2 \\ p_a & \text{if } |a| = m/2\,, \\ 1 & \text{if } |a| > m/2 \end{cases} \qquad \text{where } |a| \text{ is the Hamming weight of } a.$$

We are now ready to define our function $f\colon (\{0,1\}^{m\times r})^m \times \{0,1\}^{2^m} \to \{0,1\}$ by (see also Figure 1)

$$f(A^1,\dots,A^m,p) := \textsc{mAddr}((\textsc{Tribes}(A^1),\dots,\textsc{Tribes}(A^m)),p),$$

Note that input size of f is $n := m^2 r + 2^m$. The following constructs a narrow DNF for f.

[1] The mistake is acknowledged on the author's homepage. An implication of this result would have been to show the separation $\Pi_2\mathsf{P} \neq \mathsf{P}^{\mathsf{NP}}$ relative to a random oracle. That result however follows (using a function different from surjectivity) from the more recent result that PH is infinite in the random oracle model [15].

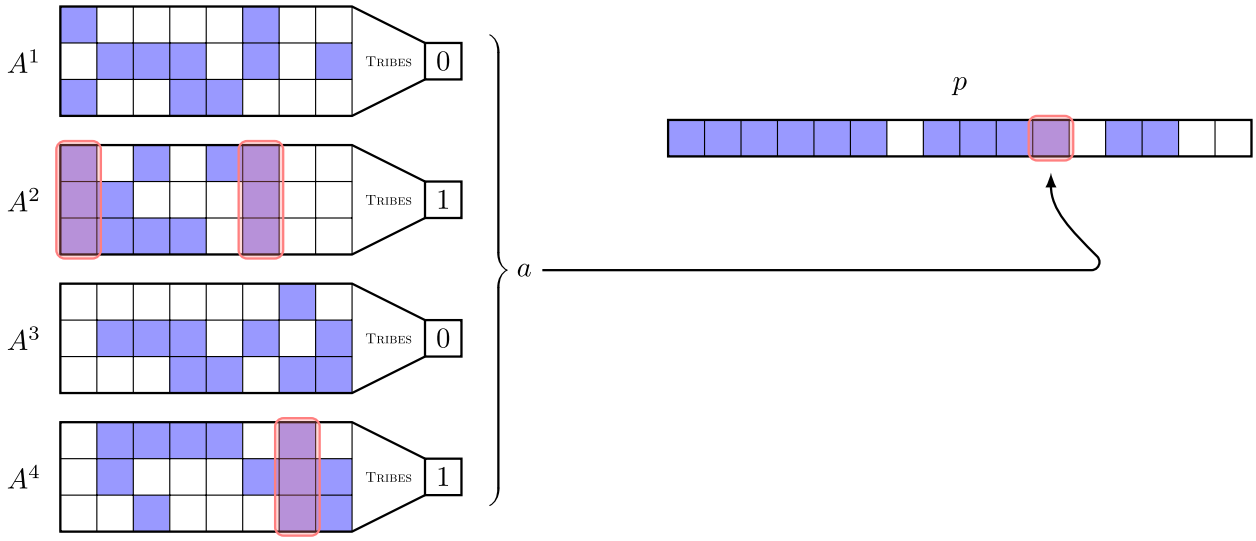


Figure 1: Illustration of $\textsc{mAddr}(\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^4), p)$. Blue cells correspond to 1-input bits, white cells correspond to 0-input bits. The address $a = (\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^4))$ is $(0, 1, 0, 1)$, so it satisfies $|a| = 4/2$. Hence the function outputs $p_{\mathrm{int}(a)} = p_{11} = 1$.

Claim 4.
There exists a DNF F of width $O(\log^2 n)$ that computes f.

Proof.
A DNF is commonly viewed as a collection of 1-*certificates*: f is computable by a k-DNF iff for each point $x \in f^{-1}(1)$ there exists a certificate comprised of a subset of input bits $I \subseteq [n]$ of size k and $\alpha \in \{0,1\}^I$ such that $x'_I = \alpha$ implies $f(x') = 1$. Hence it is enough to provide a certificate of width $O(m^2) = O(\log^2 n)$ for each 1-input of f. Consider a 1-input $x = (A^1, \ldots, A^m, p)$ and let $a := (\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^m))$. If $|a| > m/2$, a 1-certificate is simply a set of matrices $H \subseteq [m]$ of size $|H| = m/2+1$ together with a 1^m-column in each of those matrices. Such certificates fix $(m/2+1) \cdot m$ variables. A similar idea can certify 1-inputs with $|a| = m/2$, at the cost of adding the corresponding bit p_a. $\triangleleft$

We now define $\mathcal{D}$ as a distribution of the random variable $\boldsymbol{x}$ defined below.

Definition 5.
Let $\boldsymbol{x} = (\boldsymbol{A}^1, \boldsymbol{A}^2, \ldots, \boldsymbol{A}^m, \boldsymbol{p})$ *over* $\{0,1\}^n$ *be sampled as follows:*

1. *Sample* $\boldsymbol{A}^i \sim \{0,1\}^{m \times r}$ *uniformly and independently for each* $i \in [m]$.
2. *Sample* $\boldsymbol{p} \sim \{0,1\}^{2^m}$ *uniformly and independently.*
3. *Let* $\boldsymbol{a} = \big(\textsc{Tribes}(\boldsymbol{A}^1), \ldots, \textsc{Tribes}(\boldsymbol{A}^m)\big)$*; If* $|\boldsymbol{a}| = m/2$*, fix* $\boldsymbol{p}_{\boldsymbol{a}} = 1$.

We show that $\mathcal{D}$ is $(n^{-1/5}, n^{1/5})$-independent, yet f distinguishes $\mathcal{D}$ from the uniform distribution, which together with Claim 4 implies the following weaker version of Theorem 1:

Lemma 6.
The distribution $\mathcal{D}$ *as in Definition 5 is* $(n^{-1/5}, n^{1/5})$*-independent, but there is an* $O(\log^2 n)$-DNF F *such that* $\Pr_{\boldsymbol{x} \sim \mathcal{D}}[F(\boldsymbol{x}) = 1] - \Pr_{\boldsymbol{x} \sim \{0,1\}^n}[F(\boldsymbol{x}) = 1] = \Omega(\log^{-1/2} n)$.

We reduce the proof of Lemma 6 to the following two lemmas:

Lemma 7.
$\Pr_{\boldsymbol{x} \sim \mathcal{D}}[f(\boldsymbol{x}) = 1] - \Pr_{\boldsymbol{x} \sim \{0,1\}^n}[f(\boldsymbol{x}) = 1] = \Omega\big(1/\sqrt{m}\big)$*, where* m *is as in Definition 5.*

Lemma 8.
The distribution $\mathcal{D}$ *as in Definition 5 is* (ε, k)*-independent for* $k \le 2^m$*,* $\varepsilon \le k 2^{-m/2+1}$.

Proof of Lemma 6.
$\mathcal{D}$ is distributed over $\{0,1\}^n$ where $n := m^2\lceil 2^m \ln 2 \rceil + 2^m$. Apply Lemma 8 with $k = n^{1/5}$ and $\varepsilon = n^{1/5} 2^{-m/2+1} \ll n^{-1/5}$. Then by Lemma 7 and Claim 4 there exists $O(m^2) = O(\log^2 n)$-DNF that $\Omega(1/\sqrt{m}) = \Omega(1/\sqrt{\log n})$-distinguishes $\mathcal{D}$ from $\mathcal{U}$, where the latter is uniformly distributed over $\{0,1\}^n$. $\blacktriangleleft$

2.2 Proof of Lemma 7

Let $\boldsymbol{x} := (\boldsymbol{A}^1, \ldots, \boldsymbol{A}^m, \boldsymbol{p}) \sim \mathcal{D}$ and $\boldsymbol{y} \sim \{0,1\}^n$. Since the matrices $\boldsymbol{A}^i$ are uniformly generated, it is possible to couple $\boldsymbol{x}$ and $\boldsymbol{y}$ by defining $\boldsymbol{y} := (\boldsymbol{A}^1, \ldots, \boldsymbol{A}^m, \boldsymbol{p}')$ where $\boldsymbol{p}' \sim \{0,1\}^{2^m}$. Note that the address part of each input coincides and in particular, they share the event $E :=$ "$|\boldsymbol{a}| = m/2$".

Observe that $\Pr[f(\boldsymbol{x}) = 1 \mid \neg E] = \Pr[f(\boldsymbol{y}) = 1 \mid \neg E]$ by the definition of $\mathcal{D}$: if $|\boldsymbol{a}| \neq m/2$ then Step (3) is not reached in the definition of $\mathcal{D}$ and $\boldsymbol{p}$ is uniform. On the other hand we have $\Pr[F(\boldsymbol{x}) = 1 \mid E] = 1$. Indeed, if E holds, we have $F(\boldsymbol{x}) = \boldsymbol{p}_{\boldsymbol{a}} = 1$ by the definition of f and $\mathcal{D}$.

$$\begin{aligned}\Pr[F(\boldsymbol{y}) = 1 \mid E] &= \Pr[\boldsymbol{p}'_{\boldsymbol{a}} = 1 \mid E] \\ &= \sum_{a \in \{0,1\}^m} \Pr[\boldsymbol{p}'_a = 1 \mid E \wedge \boldsymbol{a} = a] \Pr[\boldsymbol{a} = a \mid E] \\ &= \sum_{a \in \{0,1\}^m} \Pr[\boldsymbol{p}'_a = 1] \Pr[\boldsymbol{a} = a \mid E] = \frac{1}{2}.\end{aligned}$$

Thus, $\Pr[f(\boldsymbol{x}) = 1] - \Pr[f(\boldsymbol{y}) = 1] = \Pr[E]/2$ and so it remains to bound $\Pr[E]$. For that, we need the following simple fact:

Lemma 9.
Let $\boldsymbol{x}$ distributed over $\{0,1\}^n$ according to a product distribution such that $|\Pr[\boldsymbol{x}_i = 1] - 1/2| \leq \varepsilon$. Then $\Delta(\boldsymbol{x}, \boldsymbol{u}) := \max_{E \subseteq \{0,1\}^n} \big|\Pr[\boldsymbol{x} \in E] - \Pr[\boldsymbol{u} \in E]\big| \leq 2n\varepsilon$, where $\boldsymbol{u} \sim \{0,1\}^n$.

Proof.
Let us couple $\boldsymbol{x}$ with $\boldsymbol{u}$ as follows: suppose $\Pr[\boldsymbol{x}_i = 1] = 1/2 + p$. We then set $\boldsymbol{u}_i := \boldsymbol{x}_i$ with probability $1/(1 + 2|p|)$ and $\boldsymbol{u}_i := [\![p > 0]\!] := (1$ if $p > 0$ otherwise $0)$ with probability $1 - 1/(1 + 2|p|) \leq 2|p| \leq 2\varepsilon$. Then $\Pr[\boldsymbol{U}_i = 1] = (1/2 + p)/(1 + 2|p|) + [\![p > 0]\!](1 - 1/(1 + 2|p|)) = 1/2$, so $\boldsymbol{u}$ is indeed uniformly distributed. Then $\Pr[\boldsymbol{x} \neq \boldsymbol{u}] \leq 2n\varepsilon$, so $\Delta(\boldsymbol{x}, \boldsymbol{u}) \leq 2n\varepsilon$. ◀

Note that each bit $\boldsymbol{a}_i$ is close to being balanced:

$$\Pr[\boldsymbol{a}_i = 1] = 1 - (1 - 2^{-m})^r = 1 - (1/e + \Theta(2^{-m}))^{\ln 2} = 1/2 + \Theta(2^{-m}).$$

As all $\boldsymbol{a}_i$ are independent, we can use Lemma 9 to get sharp bounds on their sum being exactly $m/2$: $\Pr[E] \geq \Pr_{\boldsymbol{x} \sim \{0,1\}^m}[|\boldsymbol{x}| = m/2] - \Theta(m \cdot 2^{-m}) = \Omega(1/\sqrt{m})$.

2.3 Proof of Lemma 8

We need to show that for every $I \subseteq [n]$ of size k and for every $\alpha \in \{0,1\}^I$ we have $(1 - \varepsilon) \cdot 2^{-k} \leq \Pr[\boldsymbol{x}_I = \alpha] \leq (1 + \varepsilon) \cdot 2^{-k}$. We now classify the bits of I and α. Let $I_i \subseteq [m] \times [r]$ for $i \in [m]$ be the set of bits of $\boldsymbol{A}^i$ in I. Let $J \subseteq \{0,1\}^m$ be the set of bit indices of $\boldsymbol{p}$ that belong to I (we identify the indices with their bit representations). Let $\alpha^i \in \{0,1\}^{I_i}$ and $\beta \in \{0,1\}^J$ be the corresponding parts of α.

Since $\boldsymbol{A}^1, \ldots, \boldsymbol{A}^m$ are uniformly distributed it suffices to show that

$$(1 - \varepsilon)2^{-|J|} \leq \Pr[\boldsymbol{p}_J = \beta \mid \forall i \in [m] : \boldsymbol{A}^i_{I_i} = \alpha^i] \leq (1 + \varepsilon)2^{-|J|}.$$

Let $J^{m/2} := \{s \in J \mid |s| = m/2\}$. Intuitively the only non-uniformity in $\boldsymbol{x}_I$ is introduced when $\boldsymbol{a} \in J^{m/2}$ as this is the only case where $\boldsymbol{p}$ is changed from uniform. We make this intuition precise in the following claim.

Claim 10.
For any event E that is a function of $\boldsymbol{A}^1, \ldots, \boldsymbol{A}^m$ we have

$$(1 - \Pr[\boldsymbol{a} \in J^{m/2} \mid E])2^{-|J|} \leq \Pr[\boldsymbol{p}_J = \beta \mid E] \leq (1 + \Pr[\boldsymbol{a} \in J^{m/2} \mid E])2^{-|J|}.$$

Proof.
Let $J_i := \{j \in J^{m/2} \mid \beta_j = i\}$ for $i \in \{0,1\}$. By the total probability law we get

$$\begin{aligned}\Pr[\boldsymbol{p}_J = \beta \mid E] = {} & \Pr[\boldsymbol{p}_J = \beta \mid E \wedge \boldsymbol{a} \in J_0] \Pr[\boldsymbol{a} \in J_0 \mid E] \\ & + \Pr[\boldsymbol{p}_J = \beta \mid E \wedge \boldsymbol{a} \in J_1] \Pr[\boldsymbol{a} \in J_1 \mid E] \\ & + 2^{-|J|} \Pr[\boldsymbol{a} \notin J^{m/2} \mid E] \\ = {} & 0 + 2^{-(|J|-1)} \Pr[\boldsymbol{a} \in J_1 \mid E] + 2^{-|J|} \Pr[\boldsymbol{a} \notin J^{m/2} \mid E] \\ = {} & 2^{-|J|}(\Pr[\boldsymbol{a} \notin J^{m/2} \mid E] + 2 \Pr[\boldsymbol{a} \in J_1 \mid E]).\end{aligned}$$

In (2) and (3) we use that given $\boldsymbol{a}$ the event E is independent from $\boldsymbol{p}$. Since (4) is minimized when $J_1 = \emptyset$ and maximized when $J_1 = J^{m/2}$, we have the claim. ◁

Now let E be the event "$\forall i \in [m]\colon \boldsymbol{A}^i_{I_i} = \alpha^i$". Let us compute $\Pr[\boldsymbol{a} = s \mid E]$ for $s \in \{0,1\}^m$. Since $s \in J^{m/2}$ we have $|s| = m/2$, wlog let $s = 0^{m/2}1^{m/2}$. Since the bits of $\boldsymbol{a}$ denoted by $\boldsymbol{a}_1, \dots, \boldsymbol{a}_m$ are independent and E is a conjunction of independent events we have

$$\begin{aligned}\Pr[\boldsymbol{a} = s \mid E] &= \prod_{\ell \in [m/2]} \Pr[\boldsymbol{a}_\ell = 0 \mid E] \cdot \prod_{\ell \in [m] \setminus [m/2]} \Pr[\boldsymbol{a}_\ell = 1 \mid E] \\ &\le \prod_{\ell \in [m/2]} \Pr[\boldsymbol{a}_\ell = 0 \mid \boldsymbol{A}^\ell_{I_\ell} = \alpha^\ell]\end{aligned}$$

Let us fix $\ell \in [m/2]$ and bound $\Pr[\boldsymbol{a}_\ell = 0 \mid \boldsymbol{A}^\ell_{I_\ell} = \alpha_\ell]$. By definition $\boldsymbol{a}_\ell = \textsc{Tribes}(\boldsymbol{A}^\ell)$, so it equals 0 iff no column of $\boldsymbol{A}^\ell$ is all-1, in particular all columns that do not contain bits of I_ℓ must not be all-1. For each of these columns the probability that it is not all-1 is $1 - 2^{-m}$. Since there are at least $\lceil 2^m \ln 2 \rceil - |I_\ell|$ such columns we get

$$\begin{aligned}\Pr[\boldsymbol{a} = s \mid E] &\le \prod_{\ell \in [m/2]} (1 - 2^{-m})^{\lceil 2^m \ln 2 \rceil - |I_\ell|} \\ &= (1 - 2^{-m})^{m/2 \cdot \lceil 2^m \ln 2 \rceil} (1 - 2^{-m})^{-\sum_{\ell \in [m/2]} |I_\ell|} \\ &\le 2^{-m/2} (1 - 2^{-m})^{-k} \\ &\le 2^{-m/2+1}\end{aligned}$$

Thus, $\Pr[\boldsymbol{a} \in J^{m/2} \mid E] \le |J| 2^{-m/2+1} = k 2^{-m/2+1}$, so we conclude the proof by Claim 10.

2.4 Amplification

In this section we reduce Theorem 1 to Lemma 6. The construction is a simple variation of the majority vote of several instances of f. We prove that our construction indeed amplifies the distinguishing probability in the following lemma.

Lemma 11.
Suppose $\boldsymbol{x}$ is distributed over $\{0,1\}^n$ and there exists a function $g\colon \{0,1\}^n \to \{0,1\}$ such that

$$\Pr[g(\boldsymbol{x}) = 1] - \Pr_{\boldsymbol{u} \sim \{0,1\}^n}[g(\boldsymbol{u}) = 1] \ge \delta,$$

for some δ depending on n. Let $\alpha = (\Pr[g(\boldsymbol{x}) = 1] + \Pr[g(\boldsymbol{u}) = 1])/2$. Then for $t = 2/\delta^2$ we have

$$\Pr\left[\sum_{i \in [t]} g(\boldsymbol{x}_i) \ge t \cdot \alpha\right] - \Pr\left[\sum_{i \in [t]} g(\boldsymbol{u}_i) \ge t \cdot \alpha\right] \ge \Omega(1),$$

where $\boldsymbol{x}_1, \dots, \boldsymbol{x}_t$ are independent samples of $\boldsymbol{x}$ and $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t \sim \{0,1\}^n$.

Proof.
Let $p_x = \mathbb{E}[g(\boldsymbol{x})]$. Since $\mathbb{E}\left[\sum_{i \in [t]} g(\boldsymbol{x}_i)\right] = t \cdot p_x$, we have by Hoeffding inequality,

$$\Pr\left[\sum_{i \in [t]} g(\boldsymbol{x}_i) \ge \alpha t\right] = 1 - \Pr\left[\sum_{i \in [t]} g(\boldsymbol{x}_i) < \alpha t\right] \ge 1 - e^{-2t^2(p_x - \alpha)^2/t} \ge 1 - e^{-(t\delta)^2/2t}.$$

Similarly, we can conclude that $\Pr\left[\sum_{i \in [t]} g(\boldsymbol{u}_i) \ge \alpha t\right] \le e^{-(t\delta)^2/2t}$, hence,

$$\Pr\left[\sum_{i \in [t]} g(\boldsymbol{x}_i) \ge \alpha t\right] - \Pr\left[\sum_{i \in [t]} g(\boldsymbol{u}_i) \ge \alpha t\right] \ge 1 - 2e^{-t\delta^2/2}$$

With $t = 2/\delta^2$, we conclude the proof. ◀ We now need to show that a narrow DNF can check whether $\sum_{i \in [t]} f(x_i) \ge \alpha t$. In fact, this is true for any monotone function composed with a narrow DNF:

Lemma 12.
Let $f\colon \{0,1\}^n \to \{0,1\}$ be a function that can be computed by a ℓ-DNF D. Let $g\colon \{0,1\}^t \to \{0,1\}$ be a monotone function. Then $g \circ f^t(x_1, \dots, x_t) := g(f(x_1), \dots, f(x_t))$ can be computed by a $t\ell$-DNF.

Proof.
Since f can be computed by a ℓ-DNF, a 1-certificate of f is a satisfying assignment for one term of D, which has size at most ℓ. Since g is monotone we can certify that $g \circ f^t(x_1, \dots, x_t) = 1$ by giving a 1-certificate

that $D(x_i) = 1$ for every $i \in [t]$ where that is the case. Such certificate has size at most $t\ell$, which implies that $g \circ f^t$ can be computed by a $t\ell$-DNF. ◀

Finally, we need to show that independent copies of an (ε, k)-independent distribution comprise an $(O(\varepsilon t), k)$-independent distribution:

Lemma 13.
If $\mathcal{D}$ is (ε, k)-independent, then the product distribution $\mathcal{D}^t$ is $(O(\varepsilon t), k)$-independent.

Proof.
Suppose $\boldsymbol{x} \sim \mathcal{D}$. Let $\boldsymbol{x}^t \sim \mathcal{D}^t$ be t independent copies of $\boldsymbol{x}$. Fix $I \in \binom{[n \cdot t]}{k}$ and $\alpha \in \{0,1\}^I$. For every $i \in [t]$, we define I_i and α_i to be the positions of I and α respectively in $\boldsymbol{x}_i$. Then,

$$\Pr[\boldsymbol{x}^t_I = \alpha] = \prod_{i \in [t]} \Pr[(\boldsymbol{x}_i)_{I_i} = \alpha_i] = \prod_{i \in [t]} \Pr[\boldsymbol{x}_{I_i} = \alpha_i].$$

Since $\boldsymbol{x}$ is (ε, k)-independent, for every $i \in [t]$, $(1-\varepsilon) \cdot 2^{-|I_i|} \leq \Pr[\boldsymbol{x}_{I_i} = \alpha_i] \leq (1+\varepsilon) \cdot 2^{-|I_i|}$. Hence, for small enough ε:

$$(1 - 2t\varepsilon) \cdot 2^{-k} \leq 2^{-\sum_{i \in [t]} |I_i|} \cdot (1-\varepsilon)^t \leq \Pr[\boldsymbol{x}^t_I = \alpha] \leq 2^{-\sum_{i \in [t]} |I_i|} \cdot (1+\varepsilon)^t \leq 2^{-k} \cdot (1 + 2t\varepsilon).$$

◀

Proof of Theorem 1.
Let s be a natural number to be fixed later. Let $\mathcal{D}$ be the $(s^{-1/5}, s^{1/5})$-independent distribution in Lemma 6. Let D be the $O(\log^2 s)$-DNF such that

$$\Pr_{\boldsymbol{x} \sim \mathcal{D}}[D(\boldsymbol{x}) = 1] - \Pr_{\boldsymbol{u} \sim \{0,1\}^s}[D(\boldsymbol{u}) = 1] = \Omega(1/\sqrt{\log m}).$$

From Lemma 13, for every t, $\mathcal{D}^t$ is $(O(t \cdot s^{-1/5}), s^{1/5})$-independent. By Lemma 11 for $\varphi(x_1, \ldots, x_t) := [\![\sum_{i=1}^t D(x_i) \geq \alpha t]\!] := (1$ if $\sum_{i=1}^t D(x_i) \geq \alpha t$, otherwise $0)$, when $t = O(\log s)$,

$$\Pr_{\boldsymbol{x}^t \sim \mathcal{D}^t}\left[\varphi(\boldsymbol{x}^t)\right] - \Pr_{\boldsymbol{u}^t \sim \{0,1\}^{st}}\left[\varphi(\boldsymbol{u}^t)\right] = \Omega(1).$$

Moreover, φ can be computed by a $O(t \cdot \log^2 s)$-DNF from Lemma 12. Choosing $t = O(\log s)$ and $t \cdot s = n$ we get that there exists a $(O(\log n \cdot n^{-1/5}), \Omega(n/\log n)^{1/5})$-independent distribution $\mathcal{D}^t$ over $\{0,1\}^n$ that can be $\Omega(1)$-distinguished from the uniform by a $O(\log^3 n)$-DNF, which implies the claim. ◀

2.5 Variation: Tradeoff between width and error

We finally sketch an extension of our construction that gives a tradeoff between DNF width and ε.

Theorem 14.
For any $w \geq \Omega(\log n)$ there exists a function $f_w \colon \{0,1\}^n \to \{0,1\}$ computable by a $w^{O(1)}$-DNF and an $(n^{-\Omega(w)}, n^{\Omega(1)})$-independent distribution $\mathcal{D}$ over $\{0,1\}^n$ such that

$$\Pr_{\boldsymbol{x} \sim \mathcal{D}}[f_w(\boldsymbol{x})] - \Pr_{\boldsymbol{x} \sim \{0,1\}^n}[f_w(\boldsymbol{x})] \geq \Omega(1).$$

Proof sketch.
We define a "monotone xor" of the functions Addr as follows: $g \colon (\{0,1\}^m)^w \times (\{0,1\}^{2^m})^w \to \{0,1\}$ where $g(a^1, \ldots, a^w, p^1, \ldots, p^w) := p^1_{a^1} \oplus \cdots \oplus p^w_{a^w}$ if $|a| = wm/2$, if $|a| \neq wm/2$ the value of g is 1 iff $|a| > wm/2$. The distinguisher f_w is then defined by hiding the bits of a in Tribes instances:

$$f_w(A^1, \ldots, A^{mw}, p^1, \ldots, p^w) := g(\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^{mw}), p^1, \ldots, p^w).$$

We sample $\boldsymbol{x}$ from the distribution $\mathcal{D}$ in two steps:

1.

Sample $\boldsymbol{x} = (\boldsymbol{A}^1, \ldots, \boldsymbol{A}^{mw}, \boldsymbol{p}^1, \ldots, \boldsymbol{p}^w)$ uniformly at random.

2.

If for $\boldsymbol{a} = \textsc{Tribes}^{mw}(\boldsymbol{A})$ it happens that $|\boldsymbol{a}| = wm/2$ and $g(\boldsymbol{a}, \boldsymbol{p}) = 0$, we flip a random bit among $\boldsymbol{p}^1_{\boldsymbol{a}^1}, \ldots, \boldsymbol{p}^w_{\boldsymbol{a}^w}$.

The $\Omega(1/\sqrt{mw})$-distinguishability of $\mathcal{D}$ from the uniform distribution by f_w is shown analogously to Lemma 7. Then according to Section 2.4 we increase the width of the DNF by the factor $O(mw)$ to get a $\Omega(1)$-distinguisher. The result then follows by choosing the appropriate constants in Ω and big-O.

Now we show the $(n^{-\Omega(w)}, n^{\Omega(1)})$-independence for f_w: analogously to Claim 10 one can show that to establish that $\mathcal{D}$ is $(O(\varepsilon), k)$-independent it suffices to bound $\Pr[\boldsymbol{a}^1 \in J_1 \wedge \cdots \wedge \boldsymbol{a}^w \in J_w \mid \boldsymbol{A}_I = \alpha]$ as $O(\varepsilon)$ for $J_1, \ldots, J_w \subseteq [2^m]$ and $I \subseteq ([m] \times [\lceil 2^m \ln 2 \rceil])^{mw}$ such that $|J_1| + \cdots + |J_w| + |I| \leq k$. Now for every $j = (j_1, \ldots, j_w) \in J_1 \times \cdots \times J_w$ such that $|j| = mw/2$ we have analogously to Lemma 8 $\Pr[\boldsymbol{a} = j \mid \boldsymbol{A}_I = \alpha] \leq 2^{-mw/2+w}$ as long as $|I| \leq \lceil 2^m \ln 2 \rceil$. Assuming that $|J| \leq k \leq 2^{m/4} = n^{\Omega(1)}$ we get that $\prod_{i \in [w]} |J_i| \leq 2^{mw/4}$ and therefore $\varepsilon \leq 2^{-mw/4+w} = n^{-\Omega(w)}$. ◀

3 Local couplings

3.1 Couplings fool decision lists: Proof of Theorem 3

Let $T_1, \ldots, T_M$ be the k-terms in the decision list defining f. It is sufficient to show that for $L(x) := \min\{i \in [M] \mid T_i(x) = 1\}$ we have $\Pr[L(\boldsymbol{x}) \neq L(\boldsymbol{y})] \leq 2k\varepsilon$. We show that $\Pr[L(\boldsymbol{x}) \leq L(\boldsymbol{y})]$ and $\Pr[L(\boldsymbol{y}) \leq L(\boldsymbol{x})]$ are both high and conclude the statement from that. Let us show $\Pr[L(\boldsymbol{x}) \leq L(\boldsymbol{y})] \geq 1 - k\varepsilon$ using that $(\boldsymbol{x}, \boldsymbol{y})$ is an ε-semi-coupling. Denoting $\mathrm{supp}(T_i) \subseteq [n]$ the set of input bits mentioned in the term T_i we write

$$\begin{aligned}
\Pr[L(\boldsymbol{x}) \leq L(\boldsymbol{y})] &= \sum_{i \in [N]} \Pr[L(\boldsymbol{x}) \leq i \mid L(\boldsymbol{y}) = i] \Pr[L(\boldsymbol{y}) = i] \\
&\geq \sum_{i \in [N]} \Pr[T_i(\boldsymbol{x}) = 1 \mid L(\boldsymbol{y}) = i] \Pr[L(\boldsymbol{y}) = i] \\
&\geq \sum_{i \in [N]} \Pr[\boldsymbol{x}_{\mathrm{supp}(T_i)} = \boldsymbol{y}_{\mathrm{supp}(T_i)} \mid L(\boldsymbol{y}) = i] \Pr[L(\boldsymbol{y}) = i] \\
&\geq \sum_{i \in [N]} \Pr[L(\boldsymbol{y}) = i] \left(1 - \sum_{j \in \mathrm{supp}(T_i)} \Pr[\boldsymbol{x}_j \neq \boldsymbol{y}_j \mid L(\boldsymbol{y}) = i]\right)
\end{aligned}$$

In order to conclude that $\Pr[L(\boldsymbol{x}) \leq L(\boldsymbol{y})] \geq 1 - k\varepsilon$ it suffices to show that $\Pr[\boldsymbol{x}_j \neq \boldsymbol{y}_j \mid L(\boldsymbol{y}) = i] \leq \varepsilon$. This follows from the total probability law:

$$\Pr[\boldsymbol{x}_j \neq \boldsymbol{y}_j \mid L(\boldsymbol{y}) = i] = \sum_{y \colon L(y) = i} \Pr[\boldsymbol{y} = y] \Pr[\boldsymbol{x}_j \neq \boldsymbol{y}_j \mid \boldsymbol{y} = y] \leq \varepsilon.$$

Now the same argument shows that since $(\boldsymbol{y}, \boldsymbol{x})$ is an ε-semi-coupling we have $\Pr[L(\boldsymbol{x}) \geq L(\boldsymbol{y})] \geq 1 - k\varepsilon$. We conclude Theorem 3 by the union bound.

3.2 Surjectivity fools decision lists

Aaronson [2] refuted the GLN conjecture by considering the following distribution:

Definition 15.
For every $n = m^2 2^m$, let $N = m2^m$. Define $\mathcal{D}_n$ (or simply $\mathcal{D}$ when n is clear from the context) as the distribution of $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N) \in (\{0,1\}^m)^N$ generated as follows:

1. *Sample $\boldsymbol{x}' = (\boldsymbol{x}'_1, \ldots, \boldsymbol{x}'_N) \sim (\{0,1\}^m)^N$.*
2. *Sample $\boldsymbol{y} \sim \{0,1\}^m$.*
3. *For each $i \in [N]$, let $\boldsymbol{x}_i := \boldsymbol{x}'_i$ if $\boldsymbol{x}'_i \neq \boldsymbol{y}$, otherwise $\boldsymbol{x}_i$ is sampled uniformly from $\{0,1\}^m \setminus \{\boldsymbol{y}\}$.*

Aaronson proved the following.

Theorem 16 ([2]).
For every $n = m^2 2^m$, $\mathcal{D}$ is $(k \cdot 2^{-m+1}, k)$-wise independent for all $k \leq 2^{m-1}$. Moreover, there is a depth-3 AC^0 circuit $C \colon \{0,1\}^n \to \{0,1\}$ of size $O(n^2)$ such that

$$\left| \Pr_{\boldsymbol{u} \sim \{0,1\}^n}[C(\boldsymbol{u}) = 1] - \Pr_{\boldsymbol{x} \sim \mathcal{D}}[C(\boldsymbol{x}) = 1] \right| \geq \Omega(1).$$

We prove that Aaronson's counterexample, however, cannot refute GLN conjecture for more restricted models, even decision lists.

Lemma 17.
For every $n = m^2 2^m$ and decision list $L\colon \{0,1\}^n \to \{0,1\}$ of width k,

$$\left| \Pr_{\boldsymbol{u}\sim\{0,1\}^n}[L(\boldsymbol{u}) = 1] - \Pr_{\boldsymbol{x}\sim\mathcal{D}}[L(\boldsymbol{x}) = 1] \right| \le 2k \log^2 n/n.$$

Proof.
Let $\boldsymbol{x}, \boldsymbol{x}'$ be as in Definition 15. Note that $\boldsymbol{x} \sim \mathcal{D}, \boldsymbol{x}' \sim \{0,1\}^n$. By Theorem 3, it suffices to show $\boldsymbol{x}$ is $\log^2 n/n = 2^{-m}$-coupled with $\boldsymbol{x}'$.

By definition, we need to show $(\boldsymbol{x}, \boldsymbol{x}')$ and $(\boldsymbol{x}', \boldsymbol{x})$ are 2^{-m}-semi-couplings. The former directly follows from Definition 15: for every $x' \in \{0,1\}^n$ and $i \in [N]$,

$$\Pr[\boldsymbol{x}_i \neq \boldsymbol{x}'_i \mid \boldsymbol{x}' = x'] = \Pr[\boldsymbol{x}'_i = \boldsymbol{y} \mid \boldsymbol{x}' = x'] = 2^{-m}.$$

Regarding the latter, fix any $x \in \operatorname{supp}(\mathcal{D})$, $i \in [N]$. For each $y \in \{0,1\}^m \setminus \operatorname{Im}(x)$ we have

$$\begin{aligned}
\Pr[\boldsymbol{x}'_i \neq \boldsymbol{x}_i \mid \boldsymbol{x} = x \wedge \boldsymbol{y} = y] &= \Pr[\boldsymbol{x}'_i = y \mid \boldsymbol{x} = x \wedge \boldsymbol{y} = y] \\
&= \Pr[\boldsymbol{x}'_i = y \mid \boldsymbol{x}_i = x_i \wedge \boldsymbol{y} = y] \\
&= \frac{\Pr[\boldsymbol{x}'_i = y \wedge \boldsymbol{x}_i = x_i \mid \boldsymbol{y} = y]}{\Pr[\boldsymbol{x}_i = x_i \mid \boldsymbol{y} = y]} \\
&= \frac{(2^m - 1)^{-1} 2^{-m}}{(2^m - 1)^{-1}} = 2^{-m}.
\end{aligned}$$

Crucially (5) holds since given $\boldsymbol{y} = y$ random variables $\{(\boldsymbol{x}_j, \boldsymbol{x}'_j)\}_{j\in[N]}$ are independent from each other. We conclude by the total probability law:

$$\Pr[\boldsymbol{x}'_i \neq \boldsymbol{x}_i \mid \boldsymbol{x} = x] = \sum_{y\in\{0,1\}^m\setminus\operatorname{Im}(x)} \Pr[\boldsymbol{y} = y \mid \boldsymbol{x} = x] \cdot \Pr[\boldsymbol{x}'_i \neq \boldsymbol{x}_i \mid \boldsymbol{x} = x, \boldsymbol{y} = y] = 2^{-m}.$$

◀

3.3 Semi-couplings do not fool DNFs

In this section we give an example of a semi-coupling $(\boldsymbol{x}, \boldsymbol{u})$ where $\boldsymbol{u} \sim \{0,1\}^n$ such that $\boldsymbol{x}$ can be distinguished from $\boldsymbol{u}$ by a polylogarithmic-width DNF. First, observe that we can interpret the definition of $\boldsymbol{x}$ in Definition 5 as a coupling with the uniform distribution: we sample $\boldsymbol{A}^1, \dots, \boldsymbol{A}^m, \boldsymbol{p}$ uniformly and then modify $\boldsymbol{p}$ in the location $\boldsymbol{a} = \textsc{Tribes}(\boldsymbol{A}^1), \dots, \textsc{Tribes}(\boldsymbol{A}^m)$. With $\boldsymbol{p}'$ being the state of $\boldsymbol{p}$ before the change, that defines some coupling between $\boldsymbol{x}$ and the uniformly distributed $\boldsymbol{A}^1, \dots, \boldsymbol{A}^m, \boldsymbol{p}'$. This, however, is not a semi-coupling, since if we fix $\boldsymbol{A}^1, \dots, \boldsymbol{A}^m$ to some value such that $|\boldsymbol{a}| = m/2$ and fix $\boldsymbol{p}'$ such that $\boldsymbol{p}'_{\boldsymbol{a}} = 0$, then $0 = \boldsymbol{p}'_{\boldsymbol{a}} \neq \boldsymbol{p}_{\boldsymbol{a}} = 1$ with probability 1.

We modify the distribution from Definition 5 by replacing each bit of $\mathbf{p}$ with an instance of Tribes.

Lemma 18.
There exists a $n^{-0.6}$-semi-coupling $(\boldsymbol{x}, \boldsymbol{u})$ with $\boldsymbol{u} \sim \{0,1\}^n$ and an $O(\log^2 n)$-DNF that $\Omega(\log^{-1/2} n)$-distinguishes $\boldsymbol{x}$ from $\boldsymbol{u}$.

Proof.
Consider the smallest m such that $m^2\lceil 2^m \ln 2\rceil + 2^m\lceil 2^{2m} \ln 2\rceil \ge n$. We define the coupling as follows:

1.
Sample $\boldsymbol{A} = \boldsymbol{A}^1, \dots, \boldsymbol{A}^m \sim (\{0,1\}^{m\times\lceil 2^m \ln 2\rceil})^m$ uniformly.

2.
Sample $\boldsymbol{P} = \boldsymbol{P}^1, \dots, \boldsymbol{P}^{2^m} \sim (\{0,1\}^{2m\times\lceil 2^{2m} \ln 2\rceil})^{2^m}$ uniformly.

3.
Take $\boldsymbol{Q} = \boldsymbol{P}$.

4.
Define $\boldsymbol{a} \in \{0,1\}^m$ by $\boldsymbol{a}_i = \textsc{Tribes}(\boldsymbol{A}^i)$ for each $i \in [m]$.

5.
If $|\boldsymbol{a}| = m/2$, choose $\boldsymbol{j} \sim [\lceil 2^{2m} \ln 2\rceil]$ and force $\boldsymbol{Q}^{\boldsymbol{a}}_{\ell,\boldsymbol{j}} := 1$ for each $\ell \in [2m]$.

Local coupling.

We claim that $\boldsymbol{x} := (\boldsymbol{A}, \boldsymbol{Q})$ is 2^{-2m}-semi-coupled with $\boldsymbol{u} := (\boldsymbol{A}, \boldsymbol{P})$. Fix some $A \in \mathrm{supp}(\boldsymbol{A})$ and $P \in \mathrm{supp}(\boldsymbol{P})$. Then for bits of $\boldsymbol{x}$ that correspond to $\boldsymbol{A}$ the coupling condition is trivially satisfied as these bits are shared with $\boldsymbol{u}$. The remaining bits are indexed by $a \in \{0,1\}^m$, $i \in [2m]$, $j \in [\lceil 2^{2m} \ln 2 \rceil]$, we need to bound the probability:

$$\Pr[\boldsymbol{P}^a_{i,j} \neq \boldsymbol{Q}^a_{i,j} \mid \boldsymbol{A} = A \wedge \boldsymbol{P} = P] = \Pr[\boldsymbol{Q}^a_{i,j} \neq P^a_{i,j} \mid \boldsymbol{A} = A \wedge \boldsymbol{P} = P]$$

If $|a| \neq m/2$ or $a \neq (\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^m))$, then this probability is 0 since (5) is not invoked and $\boldsymbol{P} = \boldsymbol{Q}$. If $|a| = m/2$ and $a = (\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^m))$ we have

$$\Pr[\boldsymbol{Q}^a_{i,j} \neq P^a_{i,j} \mid \boldsymbol{A} = A \wedge \boldsymbol{P} = P] \leq \Pr[\boldsymbol{j} = j] = 1/\lceil 2^{2m} \ln 2 \rceil \leq 2^{-2m} \ll n^{-0.6}.$$

Distinguishability.

We take the distinguishing function F from Lemma 7 and define the new distinguisher $F' \colon \mathrm{supp}(\boldsymbol{A}) \times \mathrm{supp}(\boldsymbol{P}) \to \{0,1\}$ as

$$F'(A^1, \ldots, A^m, P^1, \ldots, P^{2^m}) := F(A^1, \ldots, A^m, \textsc{Tribes}(P^1), \ldots, \textsc{Tribes}(P^{2^m})).$$

Let E be the event "$|\boldsymbol{a}| = m/2$". As in Lemma 7 we observe that $\Pr[\boldsymbol{P} = \boldsymbol{Q} \mid \neg E] = 1$, so $\Pr[F'(\boldsymbol{A}, \boldsymbol{P}) = 1 \mid \neg E] = \Pr[F'(\boldsymbol{A}, \boldsymbol{Q}) = 1 \mid \neg E]$. By the construction of $\boldsymbol{Q}$ and F' we have $\Pr[F'(\boldsymbol{A}, \boldsymbol{Q}) = 1 \mid E] = 1$. On the other hand

$$\begin{aligned}
\Pr[F'(\boldsymbol{A}, \boldsymbol{P}) = 1 \mid E] &= \Pr[F(\boldsymbol{A}, (\textsc{Tribes}(\boldsymbol{P}^1), \ldots, \textsc{Tribes}(\boldsymbol{P}^{2^m}))) = 1 \mid E] \\
(by Lemma 9) &\leq \Pr_{\boldsymbol{x} \sim \{0,1\}^{2^m}}[F(\boldsymbol{A}, \boldsymbol{x}) = 1 \mid E] + O(2^{-2m} \cdot 2^m) \\
(analogous to Lemma 7) &\leq 1/2 + O(2^{-m}) \leq 2/3.
\end{aligned}$$

Formally, to show the last inequality, we will do the following:

$$\begin{aligned}
\Pr_{\boldsymbol{x} \sim \{0,1\}^{2^m}}[F(\boldsymbol{A}, \boldsymbol{x}) = 1 \mid E] &= \Pr_{\boldsymbol{x} \sim \{0,1\}^{2^m}}[\boldsymbol{x}_{\boldsymbol{a}} = 1 \mid E] \\
&= \sum_{a \in \{0,1\}^m} \Pr[\boldsymbol{x}_a = 1 \mid E \wedge \boldsymbol{a} = a] \Pr[\boldsymbol{a} = a \mid E] \\
&= \sum_{a \in \{0,1\}^m} \Pr[\boldsymbol{x}_a = 1] \Pr[\boldsymbol{a} = a \mid E] = \frac{1}{2}.
\end{aligned}$$

Then as shown in Lemma 7 $\Pr[E] = \Omega(1/\sqrt{m})$. All together this gives us that F' $\Omega(1/\sqrt{m})$-distinguishes $\boldsymbol{x}$ and $\boldsymbol{u}$.

It remains to observe that the 1-certificate complexity of F' is at most $O(m^2)$: to the certificate of F in Claim 4 we add the certificate that $\textsc{Tribes}(P^j) = 1$ where $j = (\textsc{Tribes}(A^1), \ldots, \textsc{Tribes}(A^m))$. Thus there exists a DNF of width $O(m^2)$ that computes F. ◀

In order to get the $\Omega(1)$-distinguishability we follow the amplification in Section 2.4:

Theorem 19.

There exists a $1/\sqrt{n}$-semi-coupling $(\boldsymbol{x}, \boldsymbol{u})$ where $\boldsymbol{u} \sim \{0,1\}^n$ and a $O(\log^3 n)$-width DNF *that $\Omega(1)$-distinguishes $\boldsymbol{x}$ from $\boldsymbol{u}$.*

Proof.

The proof is identical to the one of Theorem 1. Take $\boldsymbol{x}'$ over $\{0,1\}^s$ that is $s^{-0.6}$-semi-coupled with $\boldsymbol{u}' \sim \{0,1\}^s$, then the random variable $\boldsymbol{x}$ comprised of $t = O(\log s)$ independent copies of $\boldsymbol{x}'$, $\boldsymbol{x} = \boldsymbol{x}'_1, \ldots, \boldsymbol{x}'_t$ is $s^{-0.6}$-semi-coupled with t independent copies of $\boldsymbol{u}'$, $\boldsymbol{u} = \boldsymbol{u}'_1, \ldots, \boldsymbol{u}'_t$. On the other hand by Lemma 12 and Lemma 11 there exists an $O(t \log^2 s) = O(\log^3 n)$-DNF that $\Omega(1)$-distinguishes $\boldsymbol{x}$ and $\boldsymbol{u}$. Since $s^{-0.6} \ll n^{-1/2}$ we get the claim. ◀

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